

NERQual: Evaluating the Robustness of Named Entity Recognition Models to Data Quality Issues

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Abstract

Transformer-based models for Named Entity Recognition (NER) achieve strong performance on clean benchmarks, yet their reliability under degraded input in real-world settings remains insufficiently understood. In practical applications, NER systems may encounter text affected by optical character recognition, automatic speech recognition, or user-generated content. We address this gap by deriving a taxonomy of data quality perturbations from a systematic literature review, consolidating fragmented prior work into a unified framework. Guided by this taxonomy, we conduct controlled stress tests evaluating six widely used transformer-based models under varying perturbation types, intensities, and training configurations. Our results show that syntactic perturbations cause the strongest performance degradation across models in this setting, while character-level processing and positional modeling are associated with different robustness patterns. Furthermore, we show that targeted training with perturbed data can recover more than half of the lost performance with minimal impact on clean-data accuracy.

1 Introduction

Named Entity Recognition (NER) is a fundamental task in Natural Language Processing (NLP) and a cornerstone of downstream applications such as information extraction and semantic search (Yadav and Bethard, 2018; Rogers et al., 2020). Transformer-based models have set strong performance standards for NER (Devlin et al., 2019; Akbik et al., 2018), especially on established benchmarks such as CoNLL-2003 (Tjong Kim Sang and de Meulder, 2003). However, such controlled benchmark settings offer only a narrow view of model reliability under degraded real-world input (Ribeiro et al., 2020; Belinkov and Bisk, 2018).

In production environments, NER systems may encounter noisy text originating from Optical

Character Recognition (OCR), Automatic Speech Recognition (ASR), or user-generated content, which can introduce spelling errors, syntactic irregularities, semantic distortions, and annotation inconsistencies (Wang et al., 2021; Belinkov and Bisk, 2018; Hamdi et al., 2023). Such data quality issues can cause severe performance degradation, with reported drops of up to one third compared to cleaner benchmark conditions (Dirkson et al., 2022; Bhadauria et al., 2024).

While robustness has attracted increasing attention in NER and related NLP tasks (Belinkov and Bisk, 2018; Lin et al., 2021; Moradi and Samwald, 2021; Bhadauria et al., 2024), existing studies typically address only limited aspects of the problem. Perturbation intensity is often operationalized and measured in different ways across studies (Li et al., 2020; Simoncini and Spanakis, 2021; Das and Paik, 2022), which renders results difficult to compare. Furthermore, previous analyses commonly focus on a restricted set of perturbation types, model families, or experimental objectives (Li et al., 2020; Bhadauria et al., 2024; Das and Paik, 2022; Simoncini and Spanakis, 2021; Lin et al., 2021; Moradi and Samwald, 2021). This fragmentation is further amplified by a lack of uniformity in evaluation settings, with studies focusing either on test-time robustness or robustness via training-time perturbations, rarely evaluating both within a single framework.

To address these limitations, we conduct a systematic experimental study that uses controlled perturbations as diagnostic probes to examine how different types and intensities of data degradation affect transformer-based NER architectures under different perturbation settings, including perturbations applied during training, at inference time, and in combination. By evaluating multiple architectures within a unified framework, our experiments enable direct comparison across perturbation types, intensity levels, and perturbation settings, while

remaining empirically limited to English CoNLL-2003. To support further research and model auditing, we provide the dataset-agnostic perturbation framework as an open-source library, designed for easy integration with other NER datasets and reproducible evaluation.¹

2 Related Work and Taxonomy

The robustness of NLP models under perturbed input has been studied through adversarial evaluation, data augmentation, challenge-set construction, and naturally occurring noise. Neural sequence labeling models are sensitive to small input perturbations, such as character-level noise or lexical substitutions (Belinkov and Bisk, 2018; Ebrahimi et al., 2018), and these issues persist for transformer-based architectures (Dirkson et al., 2022; Náplava et al., 2021; Lin et al., 2021). Although toolkits such as TextAttack, NLPAug, OpenAttack, and SeqAttack have made perturbation-based evaluation more accessible (Morris et al., 2020; Ma, 2019; Zeng et al., 2021; Simoncini and Spanakis, 2021), existing studies typically examine only limited subsets of perturbations (Simoncini and Spanakis, 2021; Li et al., 2020; Moradi and Samwald, 2021), making comparisons across experimental settings and model architectures difficult.

To obtain a more comprehensive picture of how data quality can be degraded in NER, we conducted a systematic literature review. From an initial pool of 400 publications, we identified 68 relevant studies on robustness and data quality in NLP and NER, including adversarial attacks, robustness evaluations, data augmentation, dataset analysis, and reported limitations. The systematic literature review was conducted following the guidelines proposed by Kitchenham and Charters (2007). Further methodological details are provided in Appendix A.

Our analysis revealed that several studies adopt a higher level of abstraction, grouping perturbations by their underlying linguistic characteristics rather than specific implementation details (Goyal et al., 2023; Bodapati et al., 2019; Simoncini and Spanakis, 2021; Qiao et al., 2024; Morris et al., 2020; Li et al., 2019). Following this paradigm, we categorize perturbations according to the linguistic level they affect: the orthographic surface form, the syntactic structure, or the semantic content. This hierarchical organization is summarized in Table 1.

Perturbation type	Key References
Orthographic	
General spelling	(Ai et al., 2024; Al Sharou et al., 2021; Aliakbarzadeh et al., 2024; Bagla et al., 2021; Bhadauria et al., 2024; Chen et al., 2024; Esmaail et al., 2024; Li et al., 2020; Malykh, 2019; Malykh and Lyalin, 2018; Moradi and Samwald, 2021; Moradi et al., 2021; Namysl et al., 2020; Náplava et al., 2021; Nguyen and Chen, 2023; Qiao et al., 2024)
Keystroke errors	(Araujo et al., 2020; Bagla et al., 2023, 2021; Moradi et al., 2021)
Phonetic errors	(Nguyen and Chen, 2023; Qiao et al., 2024)
Morphological	(Náplava et al., 2021)
Elongation & char repetition	(Al Sharou et al., 2021; Moradi and Samwald, 2021; Moradi et al., 2021)
OCR errors	(Ai et al., 2024; Bhadauria et al., 2024; Boros et al., 2020; Das and Paik, 2022; Ehrmann et al., 2024; Hamdi et al., 2023; Namysl et al., 2020)
Capitalization	(Ahmed et al., 2024; Al Sharou et al., 2021; Bodapati et al., 2019; Das and Paik, 2022; Esmaail et al., 2024; Moradi and Samwald, 2021; Náplava et al., 2021; Zhu et al., 2025)
Character manipulation	(Araujo et al., 2020; Ebrahimi et al., 2018; Garg and Ramakrishnan, 2020; Goyal et al., 2023; Li et al., 2020; Nguyen and Chen, 2023)
Diacritic errors	(Al Sharou et al., 2021; Náplava et al., 2021)
Syntactic	
Grammatical	(Al Sharou et al., 2021; Chen et al., 2024; Malykh, 2019; Malykh and Lyalin, 2018; Moradi and Samwald, 2021; Qiao et al., 2024; Srinivasan and Vajjala, 2023)
Word order	(Liang et al., 2024; Moradi and Samwald, 2021; Moradi et al., 2021; Náplava et al., 2021)
Sentence struct.	(Bhadauria et al., 2024; Garg and Ramakrishnan, 2020; Li et al., 2020; Shi et al., 2024; Yuan et al., 2019; Zhu et al., 2025)
Tokenization & Segmentation	(Bernier-Colborne and Vajjala, 2024; Boros et al., 2020; Hamdi et al., 2023; Szymański et al., 2023)
Punctuation	(Al Sharou et al., 2021; Esmaail et al., 2024; Goyal et al., 2023; Náplava et al., 2021)
Semantic	
Entity replacement	(Dirkson et al., 2022; Lin et al., 2021; Lothritz et al., 2023; Morris et al., 2020; Qiao et al., 2024; Srinivasan and Vajjala, 2023; Wang and Li, 2024; Yan et al., 2022)
Context word replacement	(Araujo et al., 2020; Chen et al., 2024; Das and Paik, 2022; Dirkson et al., 2022; Garg and Ramakrishnan, 2020; Jia et al., 2019; Jin et al., 2021; Li et al., 2020; Lothritz et al., 2023; Moradi and Samwald, 2021; Morris et al., 2020; Qiang et al., 2024; Wang and Li, 2024; Wu et al., 2024; Ying et al., 2022; Zeng et al., 2021; Zhu et al., 2025)
Paraphrasing	(Goyal et al., 2023; Qiang et al., 2024; Srinivasan and Vajjala, 2023; Wang et al., 2021; Zeng et al., 2021)

Table 1: Perturbation taxonomy derived from the systematic literature review, grouping text-level perturbations into orthographic, syntactic, and semantic categories with representative perturbation types and corresponding studies. The taxonomy structures the perturbation design used in the experimental evaluation.

¹<https://github.com/DFKI-NLP/NERQual>

As illustrated in Table 1, **orthographic perturbations** are the most documented category, encompassing deviations in spelling and visual representation that do not necessarily alter grammar or meaning. These include general typos (Esmail et al., 2024; Malykh and Lyalin, 2018; Náplava et al., 2021), keyboard errors (Araujo et al., 2020; Bagla et al., 2023; Moradi et al., 2021), and phonetic misspellings (Nguyen and Chen, 2023; Qiao et al., 2024). Technical noise often arises from OCR errors, such as misidentifying 'l' as '1' (Hamdi et al., 2023; Namysl et al., 2020), or diacritic inconsistencies (Al Sharou et al., 2021; Náplava et al., 2021), while regional and historical variants further challenge models (Boros et al., 2020). Capitalization issues range from inconsistent casing to full "Case Ablation" (Bodapati et al., 2019; Das and Paik, 2022). Additionally, stylistic elongations (e.g., "soooo nice") and targeted character manipulations like insertion or swapping are frequently used for synthetic robustness testing (Ebrahimi et al., 2018; Goyal et al., 2023; Li et al., 2020; Moradi and Samwald, 2021).

Syntactic perturbations concern sentence structure and grammatical correctness, testing a model's ability to process structural dependencies. Disruptions include general grammatical errors (Chen et al., 2024; Malykh and Lyalin, 2018) and word order permutations (Liang et al., 2024; Náplava et al., 2021). Structural changes, such as shortening, expansion, or word merging, further challenge contextual inference (Bhadauria et al., 2024; Li et al., 2020; Shi et al., 2024). Particularly critical for NER are tokenization and segmentation errors (Ehrmann et al., 2024; Hamdi et al., 2023; Szymański et al., 2023), where incorrect splits or lost sentence boundaries hinder entity identification and disrupt the transformer's contextual window.

Semantic perturbations disrupt meaning while maintaining a plausible surface form. A common method is replacing context words with synonyms to determine if a model recognizes an entity when familiar lexical cues change (Garg and Ramakrishnan, 2020; Li et al., 2020; Morris et al., 2020; Qiang et al., 2024). More intensive methods involve entity replacement, swapping names with others from the same category (Dirkson et al., 2022) or using distributionally mismatched terms (Lin et al., 2021; Ying et al., 2022). At the sentence level, paraphrasing is used to alter surface structure while preserving underlying semantics (Srinivasan and Vajjala, 2023; Wang et al., 2021).

3 Experimental Setup

To evaluate model robustness, we follow the established methodology of constructing challenge datasets that target specific linguistic phenomena (Belinkov and Glass, 2019). By generating multiple challenge datasets through controlled variations, we compare reference performance against targeted perturbation conditions. These perturbations are intended as diagnostic probes rather than faithful simulations of naturally occurring OCR, ASR, or user-generated noise distributions. In contrast to real-world datasets that often contain multiple, interacting sources of noise, this design allows us to isolate individual degradation types and systematically vary their intensity. Following the taxonomy introduced in Section 2, we implement a suite of orthographic, syntactic, and semantic perturbations.

To quantify how different transformer architectures adapt to these degradations, we evaluate performance across four training and testing configurations:

- **Clean-Clean:** A baseline setting where models are trained, validated, and evaluated on the original, unmodified data to establish reference performance under ideal conditions
- **Clean-Perturbed:** Models are trained and validated on clean data but evaluated on a perturbed test set, allowing measurement of immediate sensitivity to data degradation at inference time
- **Perturbed-Perturbed:** Matching perturbation types are applied during training, validation, and testing. This assesses to which extent training with degraded data can mitigate performance losses
- **Perturbed-Clean:** Models are trained and validated on perturbed data and evaluated on a clean test set to assess the model's ability to generalize to unperturbed inputs

By evaluating multiple architectures within this unified framework, our approach enables a direct comparison across different perturbation types and evaluation settings while ensuring that observed performance changes are attributable to specific degradation factors.

3.1 Data

All experiments are conducted on the CoNLL-2003 English dataset (Tjong Kim Sang and de Meulder, 2003), a widely adopted benchmark for NER. Its broad use makes it well-suited for controlled robustness experiments, as it enables direct comparison between reference and perturbed conditions while keeping the results relatable to prior NER research. At the same time, CoNLL-2003 is not free of annotation inconsistencies, and cleaned variants have been proposed in prior work (Wang et al., 2019; Rücker and Akbik, 2023). We therefore treat it as a standard reference benchmark rather than as noise-free ground truth. Our framework is implemented using the Hugging Face Datasets API (HuggingFace, 2025). We utilize the standard dataset splits to ensure reproducibility and comparability with prior work. Although this study focuses on CoNLL-2003, the framework is dataset-agnostic and can be readily applied to other NER datasets supported by the same API without changes to the perturbation or evaluation pipeline.

3.2 Models

To investigate robustness across established NER architectures, we evaluate six transformer-based models that differ in architectural design, pretraining scheme, capacity, and input representation. We use pre-trained versions of these models and fine-tune them for NER on the CoNLL-2003 dataset, reflecting a common deployment pattern for encoder-based NER systems. The selection includes:

BERT (Devlin et al., 2019) serves as the baseline architecture. We use BERT-base, which consists of 12 transformer encoder layers with 12 self-attention heads and approximately 110M parameters. The model is pretrained using masked language modeling and next sentence prediction and relies on WordPiece subword tokenization.

RoBERTa (Liu et al., 2019) follows the same encoder architecture as BERT-base (12 layers, 12 heads) but is pretrained on substantially larger and more diverse text corpora. In addition, it removes the next sentence prediction objective and applies dynamic masking, resulting in a model with approximately 125M parameters. This design isolates the effects of large-scale pretraining from architectural changes.

XLNet (Yang et al., 2019) replaces masked language modeling with permutation-based language modeling to capture bidirectional context without

masking. Comparable in size to BERT-base (12 layers, 12 heads, 110M parameters), it employs a two-stream self-attention mechanism and relative positional encodings.

DeBERTa (He et al., 2021) extends the BERT architecture by disentangling content and positional information within the attention mechanism. We use DeBERTa-base, which maintains a similar depth and attention configuration (12 layers, 12 heads) but increases model capacity to approximately 139M parameters.

DistilBERT (Sanh et al., 2019) is a compressed variant of BERT trained via knowledge distillation. It reduces the number of encoder layers from twelve to six, resulting in approximately 66M parameters while retaining the same hidden dimensionality and attention layout.

CANINE (Clark et al., 2022) differs fundamentally from the other models by operating directly on character-level input without explicit tokenization. It combines convolutional downsampling with Transformer layers to process long character sequences and contains approximately 104M parameters.

3.3 Perturbation Design and Implementation

To ensure a systematic and reproducible evaluation of data quality, we developed a custom perturbation framework. Our review of existing perturbation libraries such as TextAttack (Morris et al., 2020) and NLPAug (Ma, 2019) showed that they are not specifically designed for token-level NER and do not natively preserve BIO labels when token boundaries change. Furthermore, SeqAttack (Simoncini and Spanakis, 2021) does not implement the full range of perturbations identified in our taxonomy and lacks support for character-based models like CANINE (Clark et al., 2022). We therefore developed a dataset-agnostic NER perturbation framework that preserves BIO integrity through automated label re-alignment and remains architecture-neutral across subword and character-level models.

The profile-based framework organizes perturbations into orthographic, semantic, and syntactic classes. Each class can be configured to execute specific methods or the entire category. A dedicated control method manages candidate selection and distributes selected tokens among the active perturbation methods while enforcing selection constraints. A detailed overview of all implemented methods and examples is available in Appendix B.

We initialize the random number generators of

Python, NumPy, PyTorch, and CUDA with a shared global seed to improve reproducibility. To assess variability across runs and hyperparameter settings, each experiment is executed across five different seeds and three distinct learning rates (1e-5, 3e-5, 5e-5). Furthermore, our framework incorporates entity-protection, allowing us to isolate perturbations to either the context or the entities themselves. To create controlled stress-test conditions with increasing perturbation intensity, perturbations are applied at three levels of 10%, 20%, and 30%. This combination of variables results in a total of 2,700 training runs. While the framework is a custom development, the individual transformation methods are inspired by and adapted from established literature. Within the semantic category, we implement three primary methods for lexical and contextual replacement. Following [Chen et al. \(2024\)](#); [Srinivasan and Vajjala \(2023\)](#), we use WordNet-based synonyms and antonyms to introduce lexical variation. Inspired by [Li et al. \(2021\)](#) and [Li et al. \(2020\)](#), we use static GloVe ([Pennington et al., 2014](#)) embeddings to identify semantically similar neighbors for replacement. Finally, drawing on [Lin et al. \(2021\)](#) and [Ma \(2019\)](#), we employ Masked Language Modeling (MLM) via an independent ALBERT ([Lan et al., 2020](#)) model to generate contextually fitting perturbations.

4 Results

This section presents the experimental findings across all four evaluation settings. Performance is assessed using the entity-level micro-F1 score, necessitating an exact match for both the entity boundary and the predicted category.

4.1 Baseline

The Clean-Clean setting serves as performance ceiling, establishing a reference for model behavior under ideal conditions. As summarized in Table 2, RoBERTa and DeBERTa achieve the highest scores, followed by BERT and XLNet. DistilBERT and CANINE show slightly lower results. The low standard deviations across all models confirm the stability of the fine-tuning process.

4.2 Perturbed Test Data

This section evaluates model sensitivity to inference-time noise by testing architectures trained on clean data against perturbed test sets. As shown in Figure 1, performance correlates negatively with the proportion of modified tokens. Syn-

Model	Micro-F1 (%)	Std. Dev. σ
BERT	91.8	± 0.3
RoBERTa	92.9	± 0.2
DeBERTa	92.7	± 0.3
DistilBERT	90.3	± 0.4
CANINE	89.4	± 0.5
XLNet	91.8	± 0.3

Table 2: Baseline Performance on Clean CoNLL-2003 Test Data (clean-clean setting).

tactic perturbations cause the most severe degradation, dropping the mean F1-score below 76% at 30% intensity. This category also exhibits the highest volatility with a standard deviation of $\sigma = \pm 1.5$ at 30%. In contrast, orthographic modifications are best compensated by the models, remaining above 80% at the same noise level with lower volatility ($\sigma = \pm 0.9$). Semantic perturbations occupy an intermediate position, with a slowing degradation rate that leads semantic and orthographic performance to converge at maximum intensity.

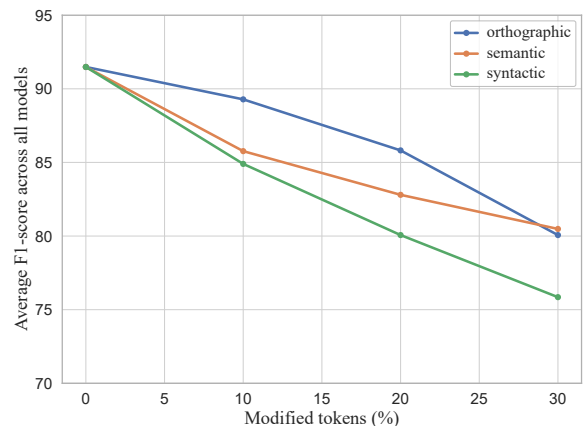


Figure 1: Mean Micro-F1 across models under increasing levels of orthographic, semantic, and syntactic test-set perturbations.

The progression of performance degradation is non-linear. Syntactic disturbances show the largest initial drop at 10% perturbation intensity. Conversely, orthographic perturbations follow a flatter trajectory initially, with a stronger decline between 20% and 30%. This pattern suggests that local surface-form changes remain easier to absorb at lower intensities, while structural perturbations affect model predictions earlier.

Detailed results per model are reported in Table 3. RoBERTa and DeBERTa achieve the strongest performance across all perturbation types, maintaining F1 scores above 85% at the initial perturbation level of 10%. This stability is reflected in

Type	Noise	BERT	RoBERTa	DeBERTa	DistilBERT	CANINE	XLNet
Baseline	0%	91.8 $\pm$ 0.3	92.9 $\pm$ 0.2	92.7 $\pm$ 0.3	90.3 $\pm$ 0.4	89.4 $\pm$ 0.5	91.8 $\pm$ 0.3
Orthographic	10%	89.7 $\pm$ 0.6	91.1 $\pm$ 0.5	90.7 $\pm$ 0.5	86.0 $\pm$ 0.8	88.6 $\pm$ 0.4	89.6 $\pm$ 0.7
	20%	85.9 $\pm$ 0.9	88.0 $\pm$ 0.6	86.9 $\pm$ 0.7	81.3 $\pm$ 1.2	86.9 $\pm$ 0.4	86.3 $\pm$ 0.8
	30%	79.8 $\pm$ 1.2	82.7 $\pm$ 0.9	82.2 $\pm$ 1.0	76.1 $\pm$ 1.6	83.9 $\pm$ 0.6	80.0 $\pm$ 1.2
Semantic	10%	86.1 $\pm$ 0.8	88.0 $\pm$ 0.7	87.7 $\pm$ 0.7	83.2 $\pm$ 1.0	83.4 $\pm$ 0.9	86.2 $\pm$ 0.8
	20%	83.3 $\pm$ 1.0	85.4 $\pm$ 0.8	84.8 $\pm$ 0.9	78.5 $\pm$ 1.2	80.9 $\pm$ 1.1	83.9 $\pm$ 1.0
	30%	81.3 $\pm$ 1.2	83.6 $\pm$ 1.0	83.7 $\pm$ 1.1	74.9 $\pm$ 1.5	78.1 $\pm$ 1.3	81.3 $\pm$ 1.2
Syntactic	10%	84.9 $\pm$ 1.0	86.7 $\pm$ 0.8	86.9 $\pm$ 0.9	81.8 $\pm$ 1.3	82.7 $\pm$ 1.1	86.4 $\pm$ 1.0
	20%	79.9 $\pm$ 1.3	82.1 $\pm$ 1.0	82.7 $\pm$ 1.1	76.6 $\pm$ 1.6	77.8 $\pm$ 1.3	81.3 $\pm$ 1.3
	30%	76.1 $\pm$ 1.5	77.7 $\pm$ 1.2	78.8 $\pm$ 1.2	72.2 $\pm$ 2.2	72.6 $\pm$ 1.6	77.7 $\pm$ 1.5

Table 3: Model performance with a perturbed test dataset (entity-level Micro-F1, %) $\pm \sigma$ over five random seeds and three different learning rates on CoNLL-2003 (clean-perturbed setting).

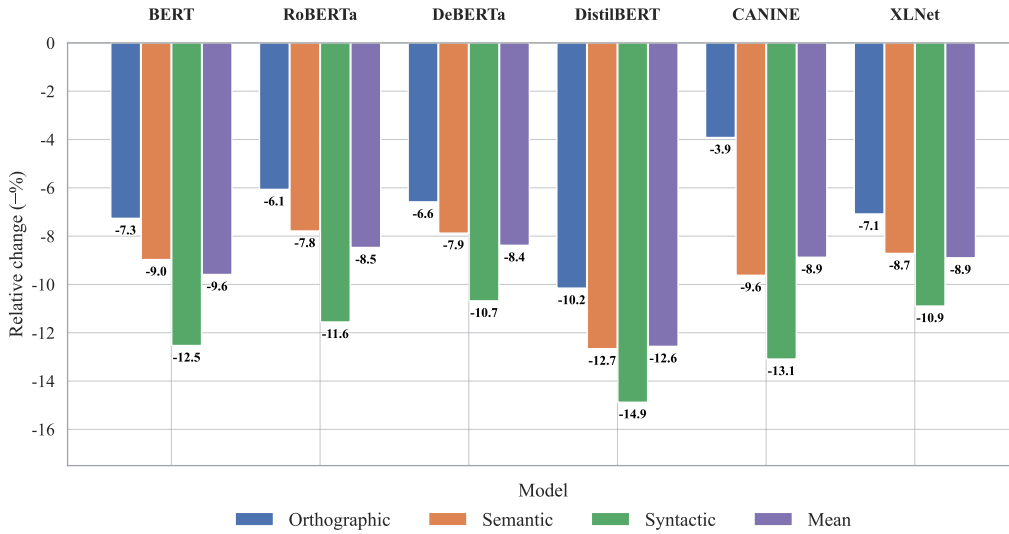


Figure 2: Relative performance change per model averaged across all intensity levels and compared to clean baselines. These averages may obscure significant failures at higher intensities; for example, stability at 10% orthographic perturbation partially offsets the sharp performance degradation seen at 30%.

their lower standard deviation ($\sigma \approx 0.8$) compared to the model average ($\sigma \approx 1.2$), indicating more stable behavior under these perturbation probes.

Model-specific robustness patterns become evident under specific degradation types. CANINE shows strong robustness to orthographic perturbations: despite a lower clean-data baseline (89.4%), it achieves the highest performance at the maximum perturbation intensity, reaching 83.9% F1 while maintaining low variability ($\sigma = 0.6$). This pattern is consistent with character-level processing being less affected by typo-induced subword segmentation errors. As illustrated in Figure 2, this corresponds to a relative drop of -3.9% for CANINE. However, this advantage is category-specific: under semantic (-9.6%) and syntactic (-13.1%) perturbations, CANINE exhibits a higher relative decline than the strongest models.

Syntactic perturbations remain the most challenging category, with even the strongest architectures dropping below 79% F1 at maximum intensity. DeBERTa achieves the highest syntactic score (78.8%), which may be related to its disentangled attention mechanism, although this interpretation cannot be isolated from differences in pretraining and capacity. Figure 2 also shows that DistilBERT has the largest average loss (-12.6%), indicating that parameter distillation may reduce the redundancy required to reconstruct contextual information from sequences where structural or semantic cues are fragmented.

4.3 Perturbed Training & Test Data

This section examines whether incorporating perturbations during training mitigates the performance degradation identified in Section 4.2.

Type	Noise	BERT	RoBERTa	DeBERTa	DistilBERT	CANINE	XLNet
Baseline	0%	91.8 $\pm$ 0.3	92.9 $\pm$ 0.2	92.7 $\pm$ 0.3	90.3 $\pm$ 0.4	89.4 $\pm$ 0.5	91.8 $\pm$ 0.3
Orthographic	10%	90.1 $\pm$ 0.4	91.6 $\pm$ 0.3	91.7 $\pm$ 0.4	87.8 $\pm$ 0.6	88.7 $\pm$ 0.5	90.7 $\pm$ 0.5
	20%	88.7 $\pm$ 0.6	90.1 $\pm$ 0.5	90.0 $\pm$ 0.5	86.5 $\pm$ 0.8	87.9 $\pm$ 0.5	89.2 $\pm$ 0.6
	30%	86.4 $\pm$ 0.8	89.0 $\pm$ 0.8	89.0 $\pm$ 0.7	84.1 $\pm$ 1.2	86.9 $\pm$ 0.6	86.6 $\pm$ 0.6
Semantic	10%	89.6 $\pm$ 0.6	90.3 $\pm$ 0.5	91.0 $\pm$ 0.5	87.7 $\pm$ 0.8	87.6 $\pm$ 0.7	89.9 $\pm$ 0.5
	20%	86.7 $\pm$ 0.8	88.3 $\pm$ 0.7	89.1 $\pm$ 0.7	84.6 $\pm$ 1.0	84.5 $\pm$ 0.9	86.9 $\pm$ 0.7
	30%	85.3 $\pm$ 0.9	87.5 $\pm$ 0.8	88.4 $\pm$ 0.7	81.9 $\pm$ 1.1	82.2 $\pm$ 1.0	85.2 $\pm$ 0.9
Syntactic	10%	88.4 $\pm$ 0.7	90.0 $\pm$ 0.6	90.4 $\pm$ 0.6	86.6 $\pm$ 0.9	85.8 $\pm$ 0.9	89.0 $\pm$ 0.8
	20%	85.3 $\pm$ 0.9	88.6 $\pm$ 0.8	89.2 $\pm$ 0.7	83.2 $\pm$ 1.2	82.5 $\pm$ 1.1	87.7 $\pm$ 1.0
	30%	83.2 $\pm$ 1.1	86.0 $\pm$ 0.9	87.5 $\pm$ 0.9	80.2 $\pm$ 1.5	80.3 $\pm$ 1.2	86.2 $\pm$ 1.0

Table 4: Model performance with perturbed training, validation, and test dataset (entity-level Micro-F1, %) $\pm \sigma$ over five random seeds and three different learning rates on CoNLL-2003 (perturbed-perturbed setting).

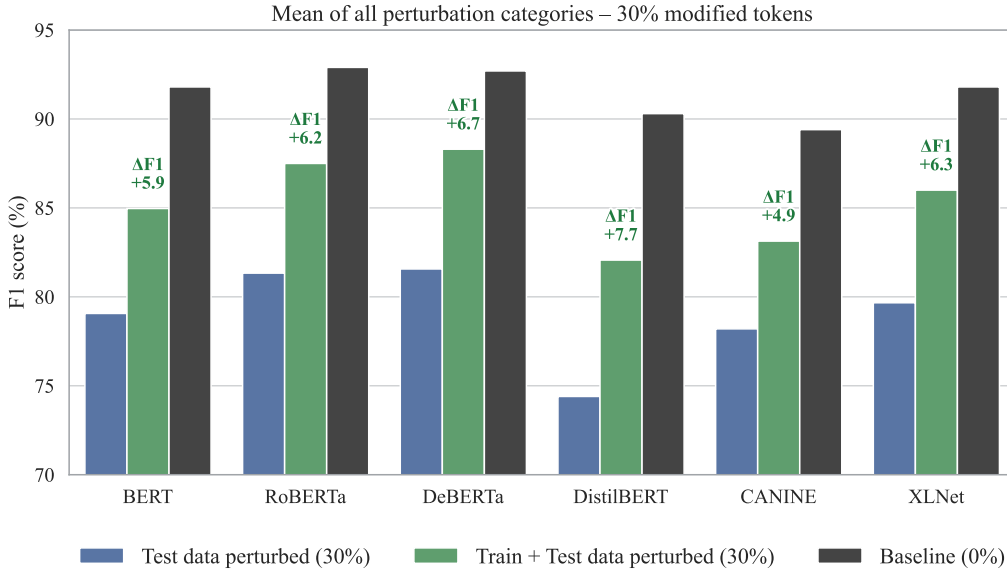


Figure 3: Mean Micro-F1 scores across all perturbation categories at 30% modified tokens, comparing clean data, perturbed test data, and perturbed training–test data for each model. $\Delta F1$ denotes the absolute improvement in Micro-F1 (in percentage points) achieved by training with perturbed data compared to perturbing the test set only.

For all architectures, Table 4 shows a noticeable performance recovery. The degradation curves follow a flatter trajectory than in the clean-perturbed setting, suggesting that training with degraded data improves stability across perturbation intensities. Average standard deviations decrease from $\sigma \approx 1.2$ to $\sigma \approx 0.8$.

To quantify this recovery, we examine the maximum load limit at 30% modified tokens in Figure 3. At this intensity, models recover on average 58.1% of the initial performance loss for orthographic, 52.1% for syntactic, and 41.9% for semantic perturbations.

The recovery differs across models and perturbation categories. At the category-mean level shown in Figure 3, DeBERTa and XLNet show high absolute gains of +6.7 and +6.3 percentage points (pp),

respectively, compared to the clean-perturbed setting. These gains indicate that both models benefit substantially from exposure to perturbed examples during training. Their strong recovery under syntactic perturbations is also consistent with the view that positional modeling mechanisms may help under structural degradation. However, this effect cannot be isolated entirely.

DistilBERT records the highest absolute gain of +7.7 pp, indicating substantial improvement when the model is trained on data from the same perturbation distribution used at test time. This result should not be interpreted as evidence that compression itself improves adaptability. Rather, it shows that a model with larger clean-perturbed losses can also exhibit larger recoverable margins under perturbation-aware training. Conversely, CANINE

shows a lower gain of +4.9 pp, which is consistent with its stronger clean-perturbed performance under orthographic perturbations and therefore less room for recovery in that category.

DeBERTa also shows the most consistent robustness gains after perturbed training. While DeBERTa and RoBERTa perform similarly on clean data, DeBERTa achieves higher gains across all perturbation categories, including +4.7 pp in the semantic domain compared to +3.9 pp for RoBERTa. This recurring advantage suggests that DeBERTa’s separation of content and positional information may support adaptation to perturbed inputs, although the present experiments do not isolate this mechanism from other model differences.

Despite this adaptability, absolute F1-scores for syntactic perturbations remain below those observed for orthographic perturbations after perturbed training. This indicates that disrupted sentence structure remains harder to handle than local spelling variation, even when the corresponding perturbation type is seen during training.

4.4 Perturbed Training Data – Clean Test Data

This section examines how training on perturbed data affects model performance on the unmodified test set. Evaluating on clean test data allows us to assess whether perturbation-aware training reduces reference performance when no input degradation is present. Comprehensive results are documented in Appendix C.

All architectures remain comparatively stable under training-phase perturbations. Unlike the larger declines observed during perturbed inference, even a 30% perturbation rate during training results in limited performance loss, with standard deviations remaining close to baseline levels. Orthographic changes have the least impact. RoBERTa remains nearly stable at 92.6% (-0.3 pp), while DeBERTa marginally exceeds its baseline with 92.8% (+0.1 pp) at the 20% level. This suggests that moderate orthographic variance may function as data augmentation, potentially preventing overfitting and improving model robustness.

Semantic and syntactic perturbations lead to slightly larger declines, but the overall losses remain limited. Even at 30% syntactic modification, RoBERTa and DeBERTa only drop to 90.5% (-2.4 pp) and 91.0% (-1.7 pp), respectively.

These results show that perturbation-aware training can improve robustness under degraded test

conditions while largely preserving performance on unmodified inputs. This suggests that, within our controlled CoNLL-2003 setting, the robustness gains observed in Section 4.3. do not come at the cost of substantial clean-data degradation.

5 Conclusion

In this paper, we evaluated the robustness of six transformer architectures for NER using a newly developed, NER-specific perturbation framework. The framework is grounded in a structured taxonomy of perturbation types and enables controlled, reproducible evaluation of data quality degradation across different intensities and training settings.

Within our English CoNLL-2003 setting, the results show clear differences in how architectures respond to controlled perturbation probes. Syntactic perturbations cause the strongest performance degradation overall, while DeBERTa and XLNet exhibit the smallest average relative declines under syntactic perturbations. This pattern is consistent with the relevance of positional modeling, although the present experiments do not isolate this factor from other model differences. Across perturbation types, RoBERTa and DeBERTa show the strongest overall stability as the proportion of modified tokens increases. CANINE shows strong robustness to orthographic perturbations, which is consistent with its character-level input representation, while DistilBERT shows larger performance drops under the evaluated perturbation conditions.

Furthermore, we show that perturbation-aware training substantially mitigates performance degradation at inference time. This strategy recovers much of the lost performance while largely preserving performance on unmodified inputs. The results therefore indicate that targeted perturbation-aware training can improve robustness in controlled stress-test settings without requiring changes to the underlying model architecture.

Future research should extend this analysis to additional datasets, domains, languages, and model classes to assess the generalizability of the observed patterns. Incorporating empirically observed real-world error distributions and validating perturbation plausibility would further clarify how controlled probes relate to naturally occurring OCR, ASR, and user-generated noise. Overall, the presented framework and findings provide a reusable basis for auditing NER robustness under imperfect input conditions.

Limitations

This study is subject to several limitations that should be considered when interpreting the results.

First, while the perturbation taxonomy is grounded in prior work, it may not capture the full spectrum of data quality issues encountered in practice. Although the taxonomy was derived from a broad literature base, certain perturbation types or variants may not be covered, which may limit the completeness of the robustness analysis.

Second, the experimental evaluation is restricted to six established encoder-based transformer architectures. While these models represent widely used and diverse design choices for token-level NER, the findings may not directly generalize to generative LLMs, multilingual models, domain-specific pretrained models, or more recent architectural variants.

Third, all experiments are conducted on the English CoNLL-2003 dataset. This benchmark enables comparability with prior NER work, but it consists of relatively homogeneous newswire text and contains documented annotation inconsistencies. Robustness patterns may differ in cleaned CoNLL variants, additional languages, or domain-specific and informal settings such as medical, legal, or social media text.

Finally, the applied perturbations are synthetically generated and uniformly distributed. They should therefore be interpreted as controlled robustness probes rather than faithful simulations of naturally occurring OCR, ASR, or user-generated noise. While the perturbation methods are linguistically motivated and grounded in prior work, we do not include a human evaluation of their realism or plausibility. Human validation and empirically observed error distributions could further clarify how controlled perturbations relate to real-world degradation.

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A Systematic Literature Review

The systematic literature review was conducted following the guidelines of Kitchenham and Charters (2007). The goal was to identify and structure prior work on robustness and data quality issues in Natural Language Processing (NLP), with a focus on Named Entity Recognition (NER) and transformer-based models.

To ensure a consistent and transparent selection process, predefined inclusion and exclusion criteria were applied throughout the review. These criteria are summarized in Table 5.

Category	Include	Exclude
Topic	Robustness or data quality issues in NLP or NER	No clear NLP focus or no substantive robustness analysis
Model Type	Transformer-based encoder architectures	Non-transformer models
Quality Issue	Explicit and reproducible perturbations	Unspecified or opaque input changes
Publication	Peer-reviewed conference or journal papers; preprints	Books or non-scientific literature
Time Frame	2017–2025	Before 2017
Language	English or German	Other languages
Availability	Full text accessible	No full-text access
Citations	More than 10 citations	10 or fewer citations

Table 5: Inclusion and exclusion criteria used in the systematic literature review.

To refine the search strategy, a pilot search evaluated candidate keywords related to robustness and noise. High-yield terms (e.g., *noise*, *robust*, *data quality*, *adversarial*) were retained, while low-yield terms were discarded.

Searches were conducted in the ACL Anthology, IEEE Xplore, ACM Digital Library, and arXiv as these were identified as the most promising sources during the pilot search. For each database, the first 100 results ranked by relevance were screened. Studies were filtered through deduplication, title and abstract screening, and full-text assessment.

The final search query was:

```
("natural language" OR "NLP" OR "named
entity recognition" OR "NER")
AND
("data quality" OR robust OR nois* OR
perturbation OR adversarial)
```

The screening process is summarized in Figure 4.

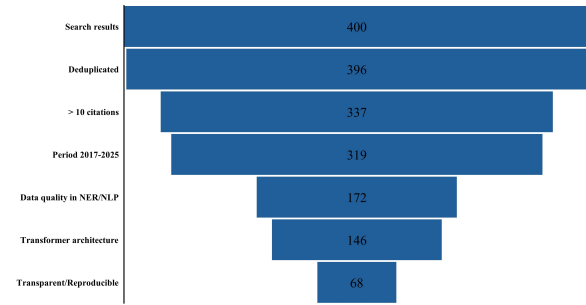


Figure 4: Study selection and screening process for the systematic literature review. Numbers indicate the remaining studies after each filtering step.

Out of the 400 initially identified studies, 68 met all inclusion criteria and were included in the final review. These studies informed the perturbation taxonomy and experimental design used in this work.

B Implemented Methods & Examples

This appendix provides qualitative examples of the perturbation methods implemented in this work, covering orthographic, syntactic, and semantic categories. The examples illustrate how controlled data quality perturbations are applied to the CoNLL-2003 corpus and correspond to the perturbation taxonomy introduced in Section 2. Orthographic, syntactic, and semantic perturbations are shown in Tables 6, 7, and 8, respectively. All perturbation implementations are publicly available in our GitHub repository <https://github.com/DFKI-NLP/NERQual>.

C Results - Perturbed Training Data

Detailed results for experiments with perturbed training and validation data evaluated on the clean test set are reported in Table 9. Across models, performance changes remain small even at higher perturbation levels.

Method	Original Text	Perturbed Text
Insertion	The company said it expects higher profits next year.	The company said it expects high h er profits next year.
Deletion	Peter Blackburn reported from London.	Peter Blackburn reported from Londo n .
Substitution	France and Germany agreed on the proposal.	France and Germany agr ee d on the proposal.
Case change	IBM said the market was stable after recent reports.	Ibm said the market was stable after recent reports.
Character swap	... would continue to support economic reforms in Europe.	... would continue to us pport economic reforms in Europe.
Diacritics	José Mourinho joined the club after talks with Chelsea officials.	Jose Mourinho joined the club after talks with Chelsea officials.
Homoglyphs	Google announced a new partnership with Apple on Monday.	Google announced a new partnership with App l e on Monday.

Table 6: Overview of implemented orthographic perturbation methods with examples from the CoNLL-2003 corpus before and after applying orthographic transformations that simulate typical spelling and typing errors. Modified tokens are highlighted in red.

Method	Original Text	Perturbed Text
Insert punctuation	The United Nations called for new peace talks in Geneva.	The United , Nations called for new peace talks in Geneva.
Delete punctuation	... from Bank of America, New York, and the United States Treasury met in London.	... from Bank of America ; New York, and the United States Treasury met in London.
Remove spaces	France and Germany signed the agreement yesterday.	Franceand Germany signed the agreement yesterday.
Split words	Microsoft announced new software on Monday.	Micro_ _ soft announced new software on Monday.
Delete words	Tony Blair met President Bush in London.	Tony Blair met President Bush in London.
Repetition	IBM reported strong results in the first quarter.	IBM IBM reported strong results in the first quarter.
Swap	Google announced a new partnership with Apple.	Google announced a new partnership Apple with.

Table 7: Overview of implemented syntactic perturbation methods with examples from the CoNLL-2003 corpus before and after applying syntactic transformation operations that simulate structural deviations in sentence organization and token segmentation. Modified tokens are highlighted in red.

Method	Original Text	Perturbed Text
WordNet-based synonym substitution	The foreign ministry’s Shen told Reuters Television ...	The foreign ministry’s Shen told Haaretz Video ...
WordNet-based antonym substitution	German first-time registrations of motor vehicles jumped ...	German unwary registrations of motor vehicles jumped ...
Embedding-based substitution (GloVe)	The meeting was attended by Tony Blair in London.	The meeting was operated by Tony Blair in London.
Contextual substitution (MLM via ALBERT)	Talks between France and Germany continued in Brussels.	Talks between France and Germany collapsed in Brussels.

Table 8: Overview of implemented semantic perturbation methods with examples from the CoNLL-2003 corpus before and after applying WordNet-, GloVe-, and MLM-based substitution techniques (here using ALBERT) to generate semantic variations. Modified tokens are highlighted in red.

Type	Noise	BERT	RoBERTa	DeBERTa	DistilBERT	CANINE	XLNet
Baseline	0%	91.8 $\pm$ 0.3	92.9 $\pm$ 0.2	92.7 $\pm$ 0.3	90.3 $\pm$ 0.4	89.4 $\pm$ 0.5	91.8 $\pm$ 0.3
Orthographic	10%	91.5 $\pm$ 0.4	92.5 $\pm$ 0.3	92.6 $\pm$ 0.3	89.9 $\pm$ 0.4	89.3 $\pm$ 0.4	91.6 $\pm$ 0.4
	20%	91.3 $\pm$ 0.3	92.7 $\pm$ 0.3	92.8 $\pm$ 0.4	89.5 $\pm$ 0.3	89.0 $\pm$ 0.5	91.6 $\pm$ 0.3
	30%	90.9 $\pm$ 0.4	92.6 $\pm$ 0.4	92.5 $\pm$ 0.4	88.8 $\pm$ 0.4	88.8 $\pm$ 0.5	91.3 $\pm$ 0.4
Semantic	10%	90.7 $\pm$ 0.4	91.7 $\pm$ 0.3	92.2 $\pm$ 0.4	89.5 $\pm$ 0.5	88.9 $\pm$ 0.5	91.2 $\pm$ 0.4
	20%	90.8 $\pm$ 0.4	91.8 $\pm$ 0.4	92.3 $\pm$ 0.4	88.9 $\pm$ 0.4	89.0 $\pm$ 0.6	91.2 $\pm$ 0.4
	30%	90.1 $\pm$ 0.3	91.6 $\pm$ 0.4	92.0 $\pm$ 0.5	88.1 $\pm$ 0.5	88.7 $\pm$ 0.6	90.8 $\pm$ 0.5
Syntactic	10%	90.1 $\pm$ 0.3	91.7 $\pm$ 0.4	91.8 $\pm$ 0.3	88.8 $\pm$ 0.5	88.5 $\pm$ 0.4	90.1 $\pm$ 0.4
	20%	89.7 $\pm$ 0.3	91.2 $\pm$ 0.3	91.3 $\pm$ 0.4	87.9 $\pm$ 0.5	87.8 $\pm$ 0.5	89.4 $\pm$ 0.5
	30%	89.1 $\pm$ 0.5	90.5 $\pm$ 0.4	91.0 $\pm$ 0.4	87.7 $\pm$ 0.6	87.1 $\pm$ 0.5	88.7 $\pm$ 0.5

Table 9: Model performance with a perturbed training & validation dataset (entity-level Micro-F1, %) $\pm \sigma$ over five random seeds and three different learning rates (1e-5, 3e-5, 5e-5) on CoNLL-2003 (perturbed-clean setting).